

Self-Supervised Blindspot Denoising for Low-SNR Stellar Spectra

VISKA WEI,^{1,2,3} ALEXANDER S. SZALAY,^{1,2} LÁSZLÓ DOBOS,^{1,4} XIAOSHENG ZHAO,¹ TAMÁS BUDAVÁRI,^{3,1,2} BALÁZS PÁL,^{5,6}
AND ROSEMARY F.G. WYSE¹

¹*Department of Physics & Astronomy, The Johns Hopkins University*

²*Department of Computer Science, The Johns Hopkins University*

³*Department of Applied Mathematics & Statistics, Johns Hopkins University*

⁴*Department of Information Systems, Eötvös Loránd University*

⁵*Department of Physics of Complex Systems, Eötvös Loránd University*

⁶*Institute for Particle and Nuclear Physics, HUN-REN Wigner Research Centre for Physics*

ABSTRACT

Low-S/N stellar spectra are unavoidable for faint survey targets, and the absorption-line measurements used for target selection, radial-velocity estimation, and abundance work are among the first quantities to degrade. Real surveys do not provide paired clean references, so fully supervised denoising inherits a simulator-to-deployment dependency. We present a one-dimensional self-supervised blindspot denoiser for stellar spectra, in which each wavelength bin is predicted from its surrounding context while the target bin is withheld from the receptive field. In a survey-relevant simulated regime of medium-resolution spectra with heteroscedastic photon, sky, and read noise, the method raises the mean linear per-spectrum pixel-level reconstruction S/N on held-out test spectra from 7.8 for the noisy input to 124 for the blindspot output on the same spectra and noise realizations. It also preserves line structure, reducing the equivalent-width error of the Ca II triplet, lowering the polarity-flip rate against the per-spectrum continuum, and improving reconstruction in the telluric O₂ A-band window. The contribution is a simulated proof-of-principle for noise-only training of stellar-spectrum denoisers; real-survey validation, deployment under correlated sky and telluric residuals, and downstream science consequences are explicitly left as future work. Code is available at <https://github.com/ViskaWei/BlindSpotDenoiser>.

Keywords: Stellar astronomy (1583) — Spectroscopy (1558) — Stellar spectral lines (1630) — Convolutional neural networks (1938)

1. INTRODUCTION

Galactic archaeology turns stellar spectra into radial velocities, atmospheric parameters, and elemental abundances that trace the formation and evolution of stellar populations. Large multiplexed surveys of resolved stars typically pre-select targets from imaging catalogs before their membership can be confirmed spectroscopically [? ?], so many faint members of the resulting samples are observed near the low-S/N limit where line measurements become fragile. Those spectra directly limit target triage, follow-up strategy, and the scientific yield of the instrument. The effective SNR of the observations is further degraded by structured night-sky emission and wavelength-dependent atmospheric absorption [? ? ?], both of which sit on top of the photon-noise floor that ultimately limits the recovery of the narrow line features carrying the astrophysically useful information [?]. A denoiser is useful in this regime only if it raises the per-spectrum reconstruction S/N without erasing those line features; line preservation is therefore the success criterion that frames the rest of this paper.

The main obstacle for learning-based denoising in this regime is not architecture but supervision. Real survey data at the magnitudes targeted here do not provide paired clean references, so a fully supervised denoiser typically depends on synthetic noisy-clean pairs and inherits a simulator-to-deployment dependency [?]. Classical template-fitting

and physical label models match observed spectra against synthetic libraries and inherit the coverage limits of those libraries [? ?], while data-driven label models such as *The Cannon* replace the synthetic library with labeled reference stars that span the target label space [?]. Each of these approaches relies in a different way on information that is not directly available from the noisy spectrum alone. A noisy-only training strategy that requires neither clean targets nor a strong external prior is therefore attractive — if it can be shown to raise reconstruction S/N without erasing the line structure that later analyses depend on.

Self-supervised denoising avoids this dependency by replacing clean training targets with statistical structure in the noisy observations themselves; even paired noisy observations of the same scene suffice to recover the underlying clean signal under zero-mean noise [?]. Blindspot training pushes this further by predicting each wavelength bin from its surrounding context alone, withholding the target bin from the receptive field so that the noisy target cannot be copied through the network [? ? ?]. The construction is well-developed for two-dimensional natural images, but stellar spectra are not images: they contain narrow absorption lines and wavelength-local correlations that do not automatically satisfy the pixelwise independence assumptions used by the classical self-supervised arguments. Whether blindspot prediction in the low-SNR stellar regime tracks those line features rather than smoothing them toward the local average is the question this paper addresses.

Our contributions are:

- A one-dimensional self-supervised blindspot denoiser for low-S/N stellar spectra, trained from noisy spectra only and evaluated against clean references only after training.
- A canonical simulated target support — continuum-normalized BOSZ-derived spectra at $R \approx 5000$, 7100–8850 Å, magnitude 20.5–22.5, and late-type stellar parameters matched to the motivating PFS-like sample.
- A direct measurement that this support lies in a small- γ regime, where model-free context predictors leave residual variance far below the per-pixel noise variance.
- Held-out comparisons against validation-tuned classical filters, noisy-PCA, and a non-deployable clean-template PCA oracle, with linear reconstruction S/N retained as the primary headline metric.
- Line-preservation diagnostics evaluated alongside aggregate reconstruction error: Ca II triplet equivalent-width error, continuum-relative polarity-flip rate, and a hard-window O₂ A-band stress diagnostic.

The contribution is a simulated proof-of-principle within this late-type, faint-magnitude support; real-survey validation, correlated residuals, radial-velocity precision, stellar-label recovery, and abundance-sensitive downstream measurements are explicitly left for future work.

2. RELATED WORK

Self-supervised denoising replaces clean training targets with statistical structure in the noisy observations themselves. [?] showed that pairs of independent noisy views of the same scene suffice to recover the clean signal under zero-mean noise, removing the need for clean references at training time but still requiring a second observation. [?] reduced this requirement further by formulating self-supervised denoising as a $\mathcal{I}$ -invariant prediction task on a single noisy frame, and [?] introduced a concrete single-image realization in which a randomly masked target pixel is predicted from its surrounding context so the noisy target cannot be copied through the receptive field. [?] converted this masking principle into a feed-forward architecture by combining branches whose receptive fields are restricted to disjoint half-spaces and then offset, so that the prediction at each output position never depends on the input at the same position. [?] extended the framework with an explicit pixel-noise model and posterior inference rather than a single point estimate, while [?] revisited the underlying $\mathcal{I}$ -invariance bound and replaced random masking with a self-consistency objective that does not require strict pixel exclusion. These methods establish the noisy-only training paradigm on two-dimensional natural images, but they leave two questions open for one-dimensional stellar spectra: whether the masked context retains enough information about the centre wavelength to support useful prediction (a regime question, addressed in Section 3.3 and quantified in Section 5.1), and whether the resulting estimator preserves the narrow absorption features that downstream stellar analysis depends on rather than smoothing them toward the local continuum (Section 5.4).

A separate line of work has applied neural denoisers directly to astronomical spectra, in all cases relying on simulator-supplied clean references. [?] train a one-dimensional denoising autoencoder on paired noisy and clean simulated stellar

spectra and show that the recovered spectra speed up template-fit initialization, at the cost of requiring synthetic clean targets at training time. [?] apply the same supervised paradigm to galaxy spectra with a noise-augmented variational autoencoder and report lower emission-line flux residuals against PCA baselines. [?] take a different route: rather than mapping noisy to clean directly, they learn a prior over clean stellar spectra from a high-S/N reference set and combine it with a generative noise model in a Bayesian denoising procedure, which still requires the high-S/N reference set during training. The common limitation across these approaches is that the supervisory signal is available only inside the simulator or inside a high-S/N reference catalogue, which is exactly the simulator-to-deployment dependency that motivates noisy-only training in this paper. Whether self-supervised blindspot training can match these reconstructions on stellar spectra, when judged jointly by aggregate reconstruction quality and by line-preservation diagnostics, has not previously been tested.

Classical spectroscopic pipelines treat low-SNR observations through fitting rather than denoising. Cross-correlation against template libraries underpins radial-velocity and redshift estimation [? ?], full-spectrum template fitting recovers stellar kinematics and population weights [? ?], and data-driven label models such as *The Cannon* and *The Payne* amortize stellar-parameter inference over learned representations of the template family [? ?]. These methods are powerful when their template assumptions hold, but at the magnitude and signal-to-noise regime targeted here they are limited by template fidelity and by the instability of the cross-correlation peak. They are therefore complementary to a denoiser rather than substitutes for it: the denoised spectra produced by the blindspot model are the kind of input on which template fitting and label inference become easier to run, and the line-preservation diagnostics in Section 5.4 are chosen to be informative for that downstream use.

The contribution of this paper sits at the intersection of these three lines, not inside any one of them. From self-supervised denoising it inherits the blindspot construction and the requirement that the centre pixel never enter the prediction of itself; from neural spectrum denoising it inherits the goal of recovering line-sensitive structure at low SNR while removing the supervision dependency that limits deployment; and with respect to classical pipelines it is positioned as upstream preprocessing rather than as a competing inference step. Two questions follow that the rest of the paper addresses directly: whether one-dimensional medium-resolution stellar spectra occupy the small- γ regime in which the masked context is informationally close to the full context (Equation 7, Section 3.3, and Section 5.1), and whether the resulting reconstruction preserves narrow absorption features rather than buying its aggregate signal-to-noise gain by mean-reversion (Section 5.4).

3. METHODS

3.1. Problem Formulation and Blindspot Objective

Let $x \in \mathbb{R}^L$ be a clean spectrum and $y \in \mathbb{R}^L$ its noisy observation:

$$y_\lambda = x_\lambda + \epsilon_\lambda, \quad \epsilon_\lambda \sim \mathcal{N}(0, \sigma_\lambda^2), \quad \lambda = 1, \dots, L, \quad (1)$$

The task is to learn a denoiser from noisy spectra only: during training, y is observed and x is not.

A blindspot predictor estimates each wavelength bin from its surrounding context with the center bin excluded:

$$\hat{x}_\lambda = f_\theta(y_{-\lambda}, \bar{\sigma}), \quad \bar{\sigma} := \sqrt{\frac{1}{L} \sum_{k=1}^L \sigma_k^2}, \quad (2)$$

Here $y_{-\lambda}$ denotes the receptive-field input after excluding y_λ , and $\hat{x}_\lambda$ is the point estimate returned by the network.

We train with a noisy-target Huber loss:

$$\mathcal{L}(\theta; y) = \frac{1}{L} \sum_{\lambda=1}^L \ell_\delta(y_\lambda - \hat{x}_\lambda), \quad \ell_\delta(r) = \begin{cases} \frac{1}{2}r^2 & |r| \leq \delta, \\ \delta(|r| - \frac{1}{2}\delta) & |r| > \delta. \end{cases} \quad (3)$$

The optimization problem is therefore

$$\theta^* \in \arg \min_{\theta} \mathbb{E}_y[\mathcal{L}(\theta; y)], \quad (4)$$

and the denoised spectrum has entries $\hat{x}_\lambda = f_{\theta^*}(y_{-\lambda}, \bar{\sigma})$. No clean spectrum is used by the training objective. Under the symmetric zero-mean per-pixel noise model of Equation 1 and a Huber breakpoint δ that is large relative to the typical fitted residual, this objective approaches the conditional mean of x_λ given $y_{-\lambda}$; asymmetric or heavy-tailed noise, or a δ comparable to or smaller than the residual scale, would shift the optimum toward an outlier-attenuated location.

3.2. Blindspot Algorithm

The forward pass enforces one invariant: the feature used to predict bin λ is never computed from y_λ . We use two shifted causal branches. The left branch processes y in wavelength order, so its feature at λ depends only on $y_{<\lambda}$. The right branch processes the wavelength-reversed spectrum and is then reversed back, so its feature at λ depends only on $y_{>\lambda}$. The prediction head receives only these two features; therefore no computational path carries y_λ into $\hat{x}_\lambda$.

Algorithm 1. Blindspot feature construction for one-dimensional spectra

Input: noisy spectrum $y \in \mathbb{R}^L$, shifted causal trunk U_θ , pointwise head g_θ .

1. Route the spectrum in the two wavelength directions: $z^L \leftarrow y$ and $z^R \leftarrow \text{rev}(y)$.
2. Apply the same shifted causal trunk: $u^L \leftarrow U_\theta(z^L)$, $\tilde{u}^R \leftarrow U_\theta(z^R)$.
3. Restore the right-branch features to the original wavelength order: $u^R \leftarrow \text{rev}(\tilde{u}^R)$, so u_λ^L sees $y_{<\lambda}$ and u_λ^R sees $y_{>\lambda}$.
4. Concatenate the two blindspot features and predict each bin: $h_\lambda \leftarrow [u_\lambda^L, u_\lambda^R]$, $\hat{x}_\lambda \leftarrow g_\theta(h_\lambda)$.

Output: $\hat{x}$, where each $\hat{x}_\lambda$ depends on $y_{<\lambda}$ and $y_{>\lambda}$ but not on y_λ .

3.3. Theory and Information Bounds

Q1: Why exclude y_λ from the receptive field, and is the context alone sufficient?—If the center pixel is visible, the noisy-target objective has a zero-loss copy solution:

$$\hat{x}_\lambda = y_\lambda \implies \mathcal{L}(\theta; y) = \frac{1}{L} \sum_{\lambda=1}^L \ell_\delta(y_\lambda - y_\lambda) = 0.$$

This solution contains no information about x_λ ; the blindspot construction removes it by excluding y_λ from the receptive field. Writing $\Omega_\lambda := (y_{-\lambda}, \bar{\sigma})$ for the blindspot context, identifiability follows from

$$y_\lambda = x_\lambda + \epsilon_\lambda, \quad \mathbb{E}[\epsilon_\lambda \mid x_\lambda, \Omega_\lambda] = 0, \quad (5)$$

which gives

$$\mathbb{E}[y_\lambda \mid \Omega_\lambda] = \mathbb{E}[x_\lambda \mid \Omega_\lambda]. \quad (6)$$

The unpredictable noise washes out under the conditional expectation, leaving the context-conditioned clean signal as the optimal regression target.

Q2: How much is lost by discarding y_λ ?—Compare three local estimators of x_λ under independent Gaussian noise: the raw pixel y_λ , the blindspot context-only estimator, and the full estimator combining both. With $\sigma^2 = \text{Var}(\epsilon_\lambda)$ and $\tau^2 := \text{Var}(x_\lambda \mid \Omega_\lambda)$, define

$$\gamma := \tau^2 / \sigma^2. \quad (7)$$

The three mean-squared errors are

$$R_{\text{raw}} = \sigma^2, \quad R_{\text{BS}} = \tau^2 = \gamma \sigma^2, \quad R_{\text{full}} = \frac{\gamma}{1 + \gamma} \sigma^2, \quad (8)$$

so the informative ratios depend only on γ :

$$\frac{R_{\text{raw}}}{R_{\text{BS}}} = \frac{1}{\gamma}, \quad \frac{R_{\text{BS}}}{R_{\text{full}}} = 1 + \gamma, \quad \frac{R_{\text{raw}}}{R_{\text{full}}} = 1 + \frac{1}{\gamma}. \quad (9)$$

As $\gamma \rightarrow 0$ the blindspot estimator tracks the full one almost exactly ($R_{\text{BS}}/R_{\text{full}} \rightarrow 1$); when $\gamma \gtrsim 1$ the center pixel carries information the context cannot supply, and blindspot training should not be expected to perform well.

Our case.—Stellar spectra are smooth functions of wavelength: continuum-normalized flux varies on scales much wider than the per-pixel sampling, and absorption lines are broadened by the instrumental line-spread function across many pixels. The neighboring bins therefore carry nearly all the information about a missing center bin, leaving very little residual variance ($\tau^2 \ll \sigma^2$), so γ should be small. We verify this premise directly on held-out clean BOSZ spectra: model-free context predictors give a conservative upper-tail scale of order 10^{-3} , with the full per-estimator measurement table given in Appendix C. We therefore use the small- γ argument as a regime statement for this simulated data set, not as an unconditional guarantee beyond it.

4. IMPLEMENTATION

Data.—The experiment uses simulated one-dimensional observations derived from the BOSZ high-resolution synthetic stellar grid [?], based on ATLAS9 atmospheres [? ?]. Each clean template is mapped to the medium-resolution setup in Table 1 and continuum-normalized before noise is added. The magnitude and stellar-parameter ranges in Table 1 match the motivating Subaru Prime Focus Spectrograph (PFS) medium-resolution stellar target sample: faint late-type spectra that remain informative despite being strongly noise dominated.

Table 1. Canonical simulated regime used in the main text. Input S/N is the per-spectrum linear ratio $\|x\|_2/\|x - y\|_2$, the same form reported as 124 for the reconstruction μ_x .

Quantity	Value
Spectral source	BOSZ synthetic stellar grid; ATLAS9 atmospheres
Wavelength range	7100–8850 Å
Resolving power	$R \approx 5000$
Magnitude bin	mag $\in [20.5, 22.5]$
Stellar-parameter support	$T_{\text{eff}} \in [3500, 6000]$ K; $\log g \in [1, 5]$; $[M/H] \in [-2.5, 0.75]$
Train / val / test spectra	200,000 / 1,000 / 1,000
Held-out noise realizations	10 per test spectrum
Input S/N (Equation 11, $X=y$)	range [2.50, 15.6], mean 7.8, median 7.8

Noise Model.—The simulator supplies a wavelength grid, clean flux x , noisy flux y , and per-pixel uncertainty vector. Its per-bin observational variance is

$$\sigma_\lambda^2 = \sigma_{\text{photon},\lambda}^2 + \sigma_{\text{sky},\lambda}^2 + \sigma_{\text{read},\lambda}^2. \quad (10)$$

The three terms encode object photon noise, structured sky background, and detector read noise after the throughput and sky model. The full pixelwise σ_λ profile is used only to draw ϵ_λ in Equation 1; it is not given to the denoiser, because it would expose location-specific simulator information. The network receives only the scalar noise summary $\bar{\sigma}$ in Equation 2.

Evaluation.—The clean reference x is withheld during training and used only for evaluation. All main-text metrics use one validation-selected model and one metric definition per column. Per-spectrum metrics are averaged across spectra within each held-out noise realization and then across realizations; reported $\pm$ values are 95% confidence intervals over the realizations in Table 1.

We use a single canonical reconstruction-S/N definition throughout. Writing $X \in \{y, \hat{x}\}$ for either the noisy input or the blindspot output evaluated against the clean reference x ,

$$\text{S/N}(X) := \frac{\|x\|_2}{\|x - X\|_2}, \quad \text{S/N}(X)|_{\text{dB}} := 10 \log_{10}[\text{S/N}(X)]^2 = 20 \log_{10} \text{S/N}(X). \quad (11)$$

Whenever reconstruction S/N is reported below, it is this same per-spectrum quantity evaluated on the held-out clean reference. The dB form differs from $10 \log_{10}[\text{Var}_\lambda(x)/\text{Var}_\lambda(x - X)]$ only by the choice between ℓ_2 -norm and per-spectrum variance for the same denominator; we report the ℓ_2 -norm form everywhere.

Equivalent width (EW) measures a spectral feature by integrating the fractional line depression relative to its continuum. For a spectrum X over a line window $[\lambda_1, \lambda_2]$,

$$\text{EW}(X) := \int_{\lambda_1}^{\lambda_2} \left(1 - \frac{F_X(\lambda)}{F_{\text{cont},X}(\lambda)} \right) d\lambda. \quad (12)$$

Because all spectra in this experiment are evaluated on the same continuum-normalized grid, $F_{\text{cont},X}(\lambda) = 1$ in the implementation and $\text{EW}_X = \int_{\lambda_1}^{\lambda_2} (1 - X(\lambda)) d\lambda$. The relative EW error against the clean reference is

$$\epsilon_{\text{EW}}(X) := \frac{|\text{EW}_X - \text{EW}_x|}{|\text{EW}_x|}. \quad (13)$$

Continuum-relative polarity measures whether each informative pixel remains on the same side of the per-spectrum continuum proxy as the clean flux. Unlike EW, which integrates a fixed line window, this statistic counts sign errors in local contrast and therefore detects line filling or contrast inversion at the pixel level. Let $m_s = \text{median}_\lambda x_s(\lambda)$ and define

$$\begin{aligned} \mathcal{P} &= \{(s, \lambda) : |x_s(\lambda) - m_s| \geq 0.02 m_s\}, \\ \mathcal{P}_- &= \{(s, \lambda) \in \mathcal{P} : x_s(\lambda) < m_s\}, \quad \mathcal{P}_+ = \{(s, \lambda) \in \mathcal{P} : x_s(\lambda) > m_s\}, \\ p_{\text{flip}}(X; \mathcal{Q}) &= \frac{1}{|\mathcal{Q}|} \sum_{(s, \lambda) \in \mathcal{Q}} \mathbf{1}[\text{sign}(X_s(\lambda) - m_s) \neq \text{sign}(x_s(\lambda) - m_s)], \quad \mathcal{Q} \in \{\mathcal{P}, \mathcal{P}_-, \mathcal{P}_+\}. \end{aligned} \quad (14)$$

The four headline metrics are:

- **Reconstruction S/N:** Equation 11, evaluated over all wavelength bins.
- **Ca ii EW error:** Equation 13 on $\mathcal{W}_c = \{\lambda : |\lambda - c| \leq 6 \text{ \AA}\}$ for $c \in \{8498, 8542, 8662\} \text{ \AA}$; reported as per-line medians over spectra and their triplet mean.
- **Polarity-flip rate:** Equation 14, reported on $\mathcal{P}$, $\mathcal{P}_-$, and $\mathcal{P}_+$.
- **Telluric S/N:** Equation 11 restricted to $\mathcal{W}_{\text{O}_2} = \{\lambda : 7590 \leq \lambda \leq 7700 \text{ \AA}\}$. Absolute values of this window-restricted statistic on the noisy input are well below unity and the linear-form S/N can become negative on the dB scale, because in the continuum-normalized representation the A-band window is dominated by sub-continuum absorption rather than continuum signal, so the numerator $\|x\|_2$ restricted to that window is small relative to the per-pixel noise scale.

Training.—We instantiate the feature construction in Algorithm 1 with a one-dimensional U-Net, scalar noise-level conditioning through $\bar{\sigma}$, and a single-head $\hat{x}$ output. The training protocol is summarized in Protocol 1; no clean reference enters the optimization.

Protocol 1. Training protocol for the reported blindspot model

Input: noisy training spectra $\{y^{(j)}\}_{j=1}^N$, simulator variances $\{(\sigma_\lambda^{(j)})^2\}$, and the blindspot model of Algorithm 1.

1. Compute the scalar noise condition for each spectrum: $\bar{\sigma}^{(j)} = \sqrt{L^{-1} \sum_\lambda (\sigma_\lambda^{(j)})^2}$; the pixelwise $\sigma_\lambda^{(j)}$ profile is not provided to the network.
2. Predict all wavelength bins with the blindspot forward pass: $\hat{x}_\lambda^{(j)} = f_\theta(y_{-\lambda}^{(j)}, \bar{\sigma}^{(j)})$.
3. Update θ against the noisy target by minimizing $L^{-1} \sum_\lambda \ell_\delta(y_\lambda^{(j)} - \hat{x}_\lambda^{(j)})$ over mini-batches and epochs.
4. Freeze the trained model after optimization.

Output: trained model f_{θ^*} and predictions $\hat{x}_\lambda = f_{\theta^*}(y_{-\lambda}, \bar{\sigma})$.

Exact model hyperparameters, evaluation-contract details, and validation-tuning rules for the non-neural reference methods are collected in Appendices A–C.

5. RESULTS

The results are organized around the burden of proof for a noisy-only blindspot denoiser. First, we test whether the simulated target-support spectra occupy the small- γ regime in which a wavelength bin is predictable from its neighbours after the bin itself is excluded. Second, we evaluate one frozen blindspot checkpoint on the held-out test split and compare it with validation-tuned classical filters and PCA references. Third, we separate the reconstruction gain from three possible failure explanations: copying the noisy input, low-pass smoothing, and low-rank projection onto the training set. Finally, we ask whether the aggregate gain preserves line-sensitive structure, and we close by stating the stability evidence and the scope of the claim. All numbers below are measured on the canonical held-out split of 1000 spectra with 10 independent noise realizations per spectrum; clean spectra are used only for evaluation.

5.1. Target-support spectra occupy a small- γ regime

Blindspot denoising is only expected to work when excluding the centre wavelength does not remove information that cannot be reconstructed from spectral context. We therefore measure this regime condition before interpreting

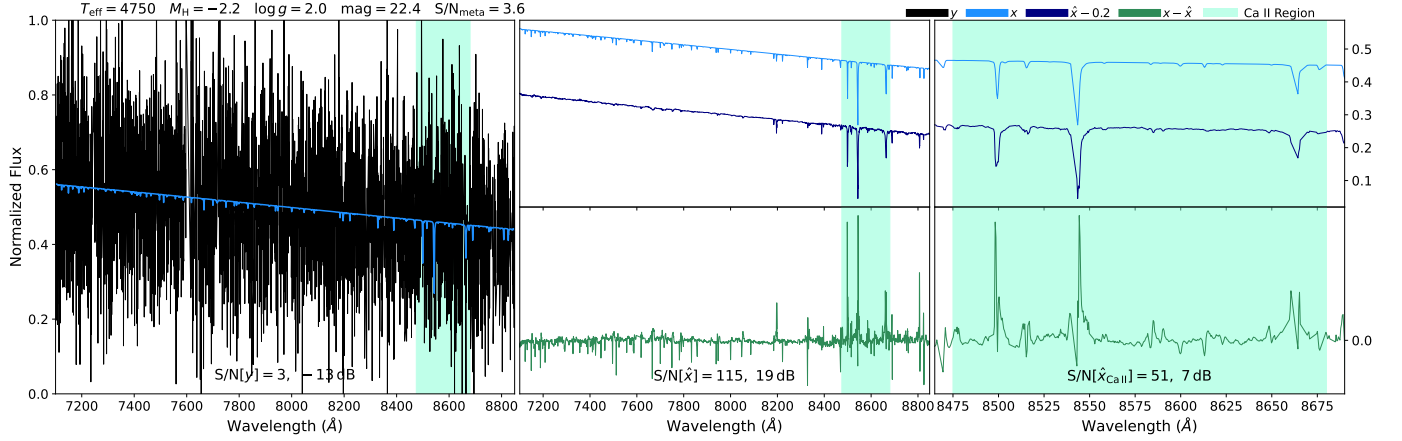


Figure 1. Representative low-S/N held-out spectrum with $T_{\text{eff}} = 4750$ K, $\log g = 2.0$, $[M/H] = -2.25$, $\text{mag} = 22.45$, and input $S/N[y] \approx 3$. **Left:** noisy input y and clean reference x on the full mask-pruned grid, with the Ca II triplet region shaded. **Middle:** clean reference x and blindspot output $\hat{x}$, offset for readability, above the residual $x - \hat{x}$; this spectrum has $S/N[\hat{x}] \approx 115$. **Right:** Ca II triplet zoom, where the line-restricted statistic is $S/N[\hat{x}_{\text{Ca II}}] \approx 51$. The output tracks the narrow Ca II cores rather than filling them toward the continuum.

any neural-network result. On held-out clean BOSZ spectra from the same target support, we estimate the context-residual ratio $\gamma = \tau^2/\sigma^2$ in Equation 7 using three model-free predictors — local cubic interpolation, context-only PCA reconstruction, and local ridge regression on a Gaussian-process surrogate — each tuned on the validation split and evaluated on the held-out test split. Table 2 reports two summary aggregations of the per-pixel ratios; the per-estimator matrix and tuning grid are deferred to Appendix C.

Table 2. Context-variance ratio $\gamma = \tau^2/\sigma^2$ on held-out clean spectra. The median-of-three row aggregates the three model-free predictors at each pixel by the per-pixel median; the worst-of-three row uses the per-pixel maximum across families and is the more conservative bound used by the small- γ argument in Section 3.3. The Ca II 8542 Å column reports the line-core peak in each aggregation; the full per-estimator and per-line table is Appendix C.

Aggregation	Median γ	95th pct. γ	Ca II 8542 Å
Median of three	2.50×10^{-5}	8.80×10^{-5}	6.94×10^{-4}
Worst of three families	2.81×10^{-4}	1.00×10^{-3}	6.70×10^{-3}

The measured residual-variance ratios are small. Even under the conservative worst-of-three aggregation, the population 95th percentile of γ is 1.00×10^{-3} ; under the median-of-three aggregation it is more than an order of magnitude smaller. The expected local peaks occur in the strongest line cores, with the largest listed value 6.70×10^{-3} at Ca II 8542 Å. The $\gamma \sim 10^{-3}$ scale used in Section 3.3 should therefore be read as an upper-tail or line-core scale, not as the population median. In this simulated target support, the wavelength context contains nearly all information needed to predict a withheld bin, while the centre-pixel noise remains unavailable to the blindspot estimator.

5.2. Held-out reconstruction over classical and low-rank baselines

Figure 1 shows the qualitative behaviour of the frozen blindspot model on a low-S/N metal-poor giant near the faint end of the magnitude bin. The input fluctuations are comparable to, or larger than, the Ca II line depths, but the blindspot output follows the clean line cores and suppresses the pixel-scale noise. Table 3 quantifies the same reconstruction problem over the full held-out test set.

The blindspot output reaches a factor of about 16 over the noisy input and more than a factor of two over the strongest noisy-only PCA reference; the clean-template PCA oracle at the same rank lies only marginally above noisy-PCA despite using information unavailable to a noisy-only deployment, so the gain is not bought by access to clean training targets and is not within reach of validation-tuned linear or wavelet baselines.

The clean-template PCA oracle is useful as a scale check but not as a competing deployment pathway: it is fit on noiseless training spectra, whereas the blindspot model is trained only against noisy observations. The comparison in Table 3 therefore separates two claims. First, the measured reconstruction gain is not reproduced by simple smoothing

Table 3. Aggregate reconstruction quality on the held-out test split. The S/N column reports the linear per-spectrum reconstruction statistic in Equation 11, averaged over the 1000 held-out spectra and 10 independent noise realizations. Classical and PCA hyperparameters are selected on the validation split. The clean-template PCA row is an oracle reference because it is fit on noiseless templates and is not deployable when only noisy survey spectra are available. The blindspot row is trained from noisy spectra only; clean spectra enter only this evaluation.

Method	Fit or training information	Status	Mean S/N
Noisy input	none	observed input	7.8
Wavelet BayesShrink	validation-tuned noisy-spectrum filter	reference	27.2
Noisy-PCA, $k=16$	PCA fit on noisy training spectra	noisy-only reference	54.4
Clean-template PCA oracle, $k=16$	PCA fit on clean templates	non-deployable oracle	56.0
Blindspot output	noisy-only blindspot training	self-supervised model	124

or by a low-dimensional linear projection of noisy spectra. Second, the gain is obtained without paired clean training targets. Figure 4 resolves the same aggregate into the joint distribution of input and output S/N over the held-out set: the bulk lies far above the identity line across the full input-S/N range, rather than being driven by a small subset of spectra.

5.3. The gain is not explained by copying, smoothing, or low-rank projection

The aggregate reconstruction number alone does not identify the mechanism. A noisy-target network could fail by copying the input if the centre pixel were visible; a smoother could raise S/N by erasing line contrast; and a low-rank model could improve reconstruction by projecting spectra onto a template subspace. We test these alternatives separately.

The copy shortcut is ruled out structurally by the blindspot construction and numerically by an integrity audit. Section 3.3 shows that a centre-visible noisy-target objective admits the zero-loss solution $\hat{x}_\lambda = y_\lambda$. The reported model removes this pathway: every prediction $\hat{x}_\lambda$ is computed from features whose receptive field excludes y_λ . A perturbation audit at three wavelength positions on held-out spectra verifies that replacing y_λ by either a cross-spectrum permutation or the per-spectrum mean changes the corresponding centre output by at most 3×10^{-7} , more than 400 times below the 10^{-4} audit threshold; the full protocol and per-perturbation table are reported in Appendix B.

The smoothing and low-rank explanations are addressed by the validation-tuned references in Table 3: although every reference improves over the noisy input, none approach the blindspot reconstruction, ruling out generic low-pass filtering and projection onto a fixed linear template subspace as explanations of the gain. The scalar noise condition $\bar{\sigma}$ used by the blindspot model also cannot carry per-pixel line information: it is one number per spectrum, broadcast across wavelength, while the full pixelwise simulator uncertainty profile is not given to the network.

Table 4 reports an ablation by training signal that verifies the copy argument empirically and bounds the self-supervised gain against a supervised oracle. Removing the blindspot and training the same architecture against noisy targets with the centre pixel visible collapses the held-out reconstruction to noisy-input scale, the trivial copy solution predicted by Section 3.3. Removing the scalar $\bar{\sigma}$ input reduces the held-out S/N from 124 to 115, while a supervised U-Net trained against clean targets reaches 157; the canonical noisy-only blindspot therefore recovers about four-fifths of the supervised oracle without paired clean targets.

Table 4. Ablation by training signal on the held-out test split. Mean S/N is the linear per-spectrum reconstruction statistic in Equation 11, averaged over the 1000 held-out spectra and 10 independent noise realizations under the same evaluation protocol as Table 3. The supervised U-Net row is an oracle reference because it requires paired clean targets unavailable to a deployed self-supervised system.

Training signal	Loss target	$\bar{\sigma}$ input	Mean S/N
Centre-visible (no blindspot)	noisy y	none	7.8
Blindspot, no $\bar{\sigma}$ input	noisy y	none	115
Blindspot + scalar $\bar{\sigma}$	noisy y	scalar $\bar{\sigma}$	124
Supervised U-Net (oracle)	clean x	none	157

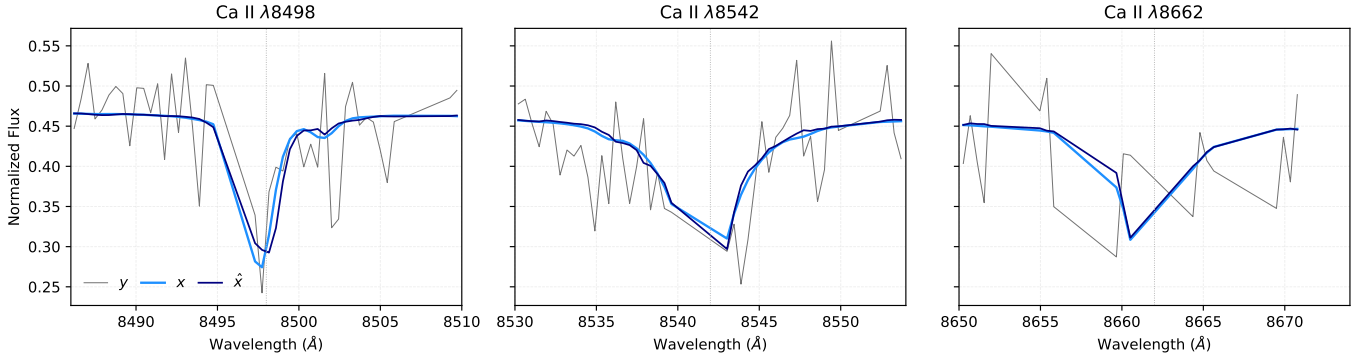


Figure 2. Ca II triplet zoom for the same representative held-out spectrum as Figure 1. The panels show the 8498, 8542, and 8662 Å lines. Each panel overlays the noisy input y , the clean reference x , and the blindspot output $\hat{x}$. The blindspot reconstruction follows the depth and width of the Ca II cores without flattening them toward the continuum.

5.4. Line-sensitive reconstruction diagnostics

A spectrum denoiser is not useful if it raises pixel-level S/N by erasing absorption features. We therefore evaluate the frozen blindspot output with diagnostics that are sensitive to line filling and contrast inversion: Ca II triplet equivalent-width error, continuum-relative polarity flips, and reconstruction in the telluric O₂ A-band window. These diagnostics are not downstream radial-velocity, abundance, or label-recovery tests; they measure whether the reconstructed spectra preserve line structure in the simulated data.

Table 5. Line-sensitive reconstruction diagnostics on the held-out test split. Equivalent-width entries are per-spectrum medians of the relative error in Equation 13. The polarity rows count sign reversals relative to the per-spectrum continuum proxy in Equation 14. The O₂ A-band rows are included as a hard-window stress diagnostic, not as the headline result.

	Noisy input y	Blindspot output $\hat{x}$	$\Delta = \hat{x} - y$
<i>Ca II triplet relative EW error</i>			
8498 Å	0.653	0.117	−0.536
8542 Å	0.338	0.092	−0.246
8662 Å	1.025	0.143	−0.882
Triplet mean	0.672	0.117	−0.555
<i>Polarity-flip rate against the per-spectrum continuum median</i>			
All non-continuum pixels	0.271	0.00290	−0.268
Below-continuum pixels ($x < m$)	0.264	0.00340	−0.260
Above-continuum pixels ($x > m$)	0.278	0.00230	−0.276
<i>Telluric O₂ A-band hard window (7590–7700 Å, 181 pixels)</i>			
Window-restricted S/N, linear	0.16	3.6	+3.4
Mean relative EW degradation	25.8	1.34	−24.5

The Ca II triplet errors decrease for all three lines. The triplet-mean relative EW error falls from 0.672 on the noisy input to 0.117 after blindspot denoising, and the largest improvement occurs for the 8662 Å line, where the noisy-input median error exceeds unity. The continuum-relative polarity-flip rate falls from 0.271 to 0.00290 over all non-continuum pixels, with comparable improvements for below-continuum and above-continuum pixels at the same fixed threshold (sensitivity to the threshold value is left to Appendix D). These polarity results are a direct check against line filling: if the model were simply pulling features toward the continuum, sign errors relative to the clean spectrum would remain common.

The O₂ A-band rows give a difficult window-specific stress test. This region is not the primary headline because it is a narrow diagnostic window rather than a full-spectrum reconstruction metric, but it is useful because the signal is weak and the absorption structure is strong. In this window, the linear reconstruction statistic rises from 0.16 to 3.6,

and the mean relative EW degradation decreases from 25.8 to 1.34. Together with the Ca II and polarity rows, this supports the interpretation that the reconstruction gain is line-preserving rather than mean-reverting.

Figure 2 shows the same conclusion visually for the Ca II triplet. The noisy trace has excursions larger than the line-core contrast, while the blindspot output follows the clean line morphology. The quantitative diagnostics in Table 5 therefore measure the behaviour seen in the representative spectrum rather than relying on it.

5.5. Repeated-noise stability and scope

The preceding results use one frozen trained model. We therefore test whether the reported output is stable to repeated noise perturbations of a fixed clean spectrum. Holding one held-out clean spectrum x fixed, we draw 10^4 independent noise realizations from the same heteroscedastic simulator, denoise each realization, and summarize the output ensemble by its mean $\langle \hat{x} \rangle$ and across-draw spread $\sigma_{\hat{x}}$.

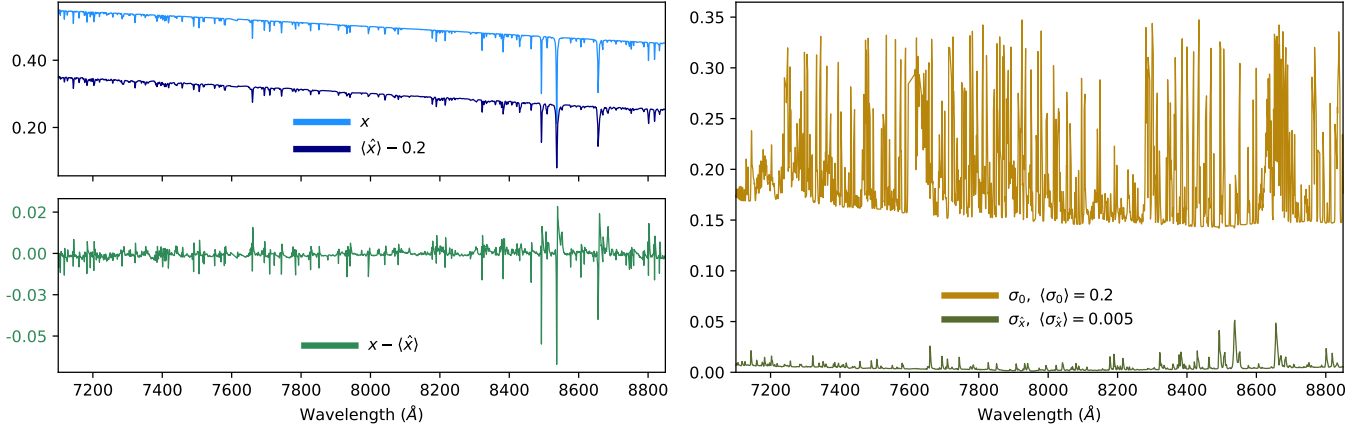


Figure 3. Fixed-spectrum repeated-noise diagnostic for the reported blindspot model. One held-out clean spectrum is perturbed by 10^4 independent noise draws and denoised repeatedly. **Top-left:** clean spectrum x and repeated-draw mean estimate $\langle \hat{x} \rangle$, shifted for readability. **Bottom-left:** residual $x - \langle \hat{x} \rangle$. **Right:** input noise scale and across-draw spread $\sigma_{\hat{x}}$ of the denoised outputs on a shared linear axis. This diagnostic measures sensitivity to noise realizations for a fixed spectrum; it is not a calibrated uncertainty estimate.

Figure 3 shows that the repeated-draw mean follows the clean spectrum and that the denoised spread remains far below the input noise scale over most wavelengths. We interpret this as a stability diagnostic for the point estimator: changing the random noise realization does not produce large changes in the denoised output for this fixed spectrum. It should not be read as posterior uncertainty calibration, as an error bar on individual line measurements, or as evidence for robustness to all training seeds.

All conclusions are conditional on the simulated target support studied here: continuum-normalized, medium-resolution, late-type BOSZ/PFS-like spectra at magnitude 20.5–22.5 with independent zero-mean per-pixel noise. The results do not validate deployment on real survey spectra, correlated sky or telluric residuals, hotter or sharper-lined spectra, fainter targets, uncertainty calibration, radial-velocity precision, stellar-parameter recovery, or abundance-sensitive downstream measurements.

6. DISCUSSION

In this simulated target support, noisy-only blindspot training raises the mean linear reconstruction S/N from 7.8 on the noisy input to 124 on held-out spectra, exceeding validation-tuned classical filters, noisy-PCA, and a non-deployable clean-template PCA oracle. The line-sensitive diagnostics in Section 5.4 show that this gain is not obtained by simply filling absorption features toward the continuum: Ca II EW errors and continuum-relative polarity flips both decrease substantially.

Scope of the result.—All measurements are obtained on continuum-normalized synthetic spectra drawn from the magnitude and stellar-parameter support of the target observed sample; the noise model is independent and zero-mean per-pixel. The reported gain therefore characterizes blindspot denoising in this survey-relevant low-SNR support, not extrapolation to fainter targets, hotter stars, or any particular real survey.

Where the method should be expected to fail.—Three failure modes are predictable from the assumptions in Section 3. First, structured residuals — imperfectly subtracted sky lines, telluric mismatch, and detector cosmetics — violate the conditional-independence assumption of Equation 1 and can drive the context-conditioned predictor toward biased reconstructions inside affected windows. Second, sharper unresolved features and hot-star spectra raise the local conditional variance τ^2 , which moves γ out of the small regime in which the blindspot estimator tracks the full estimator (Section 3.3). Third, the Huber objective in Equation 3 estimates an outlier-attenuated conditional location rather than the conditional mean unless the noise is symmetric to within the breakpoint δ and δ is large relative to the residual scale; asymmetric or heavy-tailed corruption is therefore expected to shift the optimum away from the target in Section 3.1.

Reproducibility scope.—All main numerical results are obtained from one frozen canonical checkpoint evaluated on one held-out split with ten independent noise realizations per spectrum. The repeated-noise diagnostic in Figure 3 tests sensitivity to noise draws for a fixed clean spectrum; it is not uncertainty calibration and it does not substitute for a full training-seed robustness study. The copy-shortcut path is ruled out structurally by the blindspot construction and by the integrity audit; we do not run a separate centre-visible/noisy-target reconstruction control. Independent training-seed repeats, scalar-noise-conditioning ablations, and supervised clean-target references are left outside the main claim unless evaluated under the same frozen held-out metric battery.

Real-data validation, downstream science consequences (radial-velocity precision, stellar-parameter recovery, abundance-sensitive features), and stress tests against correlated residuals are explicitly left as future work.—The present manuscript is a simulated proof-of-principle for noise-only training within the stated target support, not a survey-deployment or out-of-support extrapolation claim.

7. CONCLUSION

This study establishes a narrow but useful point: blindspot self-supervision can act as a credible denoising constraint for simulated low-S/N stellar spectra when clean targets are absent. The result is not only that the learned estimator improves a reconstruction score on withheld spectra. It is that the improvement survives the measurements that would make a denoiser scientifically suspect if it merely smoothed toward the continuum: absorption features remain usable, sign mistakes around the continuum are reduced, and a difficult atmospheric window is not treated as disposable background. In that sense the blindspot condition is doing more than regularization by architecture. It supplies an observationally plausible training signal in the regime where the target value itself cannot be trusted as input, and where paired clean spectra would normally have to be imported from a simulator.

The claim is deliberately bounded. The spectra are continuum-normalized, drawn from the same synthetic support as the benchmark, and corrupted by independent, zero-mean binwise noise. Within that setting, withholding the centre bin gives enough neighboring context for the network to recover stable spectral structure without learning the identity shortcut. Outside that setting, the same mechanism can fail for reasons that are not incidental: correlated subtraction residuals, detector artifacts, mismatched atmospheric absorption, or spectra with intrinsically higher local ambiguity can all make the context-conditioned target biased or underdetermined. The method should therefore be read as a deployment constraint with demonstrated in-support value, not as evidence that any survey spectrum can be cleaned without consequence.

The practical implication is that clean-reference scarcity need not force denoising to be fully supervised, provided the noise assumptions and spectral support are made explicit and audited. The next boundary is the behavior of this constraint under real survey residuals and its effect on downstream stellar inference.

ACKNOWLEDGMENTS

This work is supported by the generosity of Eric and Wendy Schmidt, by recommendation of the Schmidt Futures program.

Software: Astropy, NumPy, SciPy, Matplotlib, PyTorch

APPENDIX

A. EXPERIMENTAL AND REPRODUCIBILITY CONTRACT

This appendix collects the frozen experimental contract under which every number reported in Sections 5.2–5.5 is obtained: the public release that pins code and data, the canonical training and held-out split, the rule that separates validation tuning from held-out evaluation, and the software environment. Apart from the trained-checkpoint row, this contract is model-agnostic and is intended to be reusable for future denoisers evaluated on the same simulated support.

Archived release.—The trained checkpoint, evaluation script, simulator configuration, and per-baseline tuning record will be bundled into an archived public release at the time of submission; the artifacts that define the frozen evaluation contract used for every quantitative table in this paper are listed in Table 6, and each row identifies the asset that reproduces the corresponding row of a paper table or panel of a paper figure.

Table 6. Artifacts defining the frozen evaluation contract. Each entry resolves to an asset inside the archived release; the same release pins the trained checkpoint, the held-out split, and the metric implementation used for every quantitative result in this paper.

Artifact	Role	Reproduces
Trained checkpoint	frozen denoising model	Table 3, Table 5; Figs. 1, 2, 3
Held-out test split	frozen evaluation set	all held-out test rows
Metric implementation	frozen scorer	all metric columns
Validation split + tuning record	per-baseline grids	Table 3, Table 9
γ predictor record	per-estimator measurements	Table 2, Table 11
Simulator configuration	noise model	Table 1
Environment specification	software pin	all numerical results

Validation and held-out separation.—The training, validation, and held-out test sets contain 200,000, 1,000, and 1,000 spectra respectively, drawn without replacement from the same regime-restricted BOSZ library so that the marginal distributions over T_{eff} , $\log g$, $[M/H]$, and magnitude match across splits. All hyperparameter selection — the training-time validation rule for the blindspot model, the classical-baseline grids in Appendix C, and the model-free γ predictors — uses the validation split alone, with a single fixed validation noise seed. Once a hyperparameter is selected it is frozen, and every quantitative result in this paper is computed on the held-out test split with 10 independent noise realizations per spectrum drawn from a deterministic seed sequence. Per-spectrum metrics are first averaged across spectra at each realization and then across realizations; reported $\pm$ values are 95% confidence intervals over the realizations.

Software environment.—A dependency manifest and a container recipe are pinned in the archived release, so that the metric module, simulator, and inference script execute against the same software stack used to generate the reported numbers.

B. CANONICAL BLINDSPOT MODEL AND LEAKAGE AUDIT

This appendix is specific to the blindspot model and is not part of the model-agnostic evaluation contract of Appendix A. It records the canonical model architecture and training configuration that produce the held-out reconstruction in Section 5.2 and the numerical blindspot integrity audit that rules out leakage of y_λ into $\hat{x}_\lambda$ in the reported network, in support of Sections 3.2 and 5.3.

Canonical model and training configuration.—The denoiser is a one-dimensional convolutional U-Net that maps a single-channel noisy spectrum, concatenated with the broadcast scalar $\bar{\sigma}$ as a second input channel, to a single-channel reconstruction $\hat{x}$. The backbone uses nine convolutional layers with width 48 and kernel size 25, batch normalization, and residual aggregation between matching encoder and decoder levels; the output is a single-head $\hat{x}$ point estimate, with no auxiliary σ output head and no probabilistic output branch. The blindspot property is enforced by combining shifted causal convolutions on each side of the spectrum, as in Algorithm 1. Optimization uses Adam with $(\beta_1, \beta_2, \epsilon) = (0.9, 0.999, 10^{-8})$ and zero weight decay, initial learning rate 10^{-4} , and a cosine schedule decaying to 10^{-6} over the full training horizon. Training uses 200 epochs, batch size 128, and a single fixed training seed; the early-stopping

patience is 30 epochs and the reported model is selected as the validation-best by reconstruction S/N before any held-out evaluation. At each training step, 85% of wavelength bins are sampled at random and contribute to the noisy-target loss, while the remaining bins are held out from the loss for that step; this stochastic loss-mask is the standard blindspot-training mechanism for decorrelating successive predicted bins and does not modify the network input or the held-out evaluation, which is computed at every wavelength.

Table 7. Canonical blindspot model and training configuration used for all main-text and appendix results. The training-time loss-mask ratio is the random fraction of wavelength bins that contribute to the noisy-target loss at each step (the loss-mask trick); held-out evaluation is computed at every wavelength.

Quantity	Value
Architecture	1D shifted-causal blindspot U-Net
Convolutional layers	9
Width	48
Kernel size	25
Normalization	batch normalization
Output head	single-channel $\hat{x}$ (no auxiliary σ head)
Input channels	noisy spectrum + broadcast scalar $\bar{\sigma}$
Loss	Huber, $\delta = 0.03$ in normalized flux units
Training target	noisy observation
Optimizer	Adam, $(\beta_1, \beta_2, \epsilon) = (0.9, 0.999, 10^{-8})$
Weight decay	0
Initial learning rate	10^{-4}
Learning-rate schedule	cosine decay to 10^{-6}
Batch size	128
Epochs	200
Early-stopping patience	30 epochs
Training-time loss-mask ratio	0.85 (per step, stochastic)
Training seed	0 (fixed)
Validation noise seed	fixed across training
Checkpoint selection rule	validation-best reconstruction S/N

Numerical blindspot integrity.—The blindspot construction implies that $\hat{x}_\lambda$ depends on the input only through positions other than λ . We test this property numerically at three test-pixel positions, $i \in \{L/4, L/2, 3L/4\}$, on 64 held-out spectra. For each spectrum and each position we forward the network on the noisy input and on a perturbed input that modifies y_i but leaves y_j unchanged for $j \neq i$, and record the central-pixel deviation $|\hat{x}_i^{\text{perturbed}} - \hat{x}_i^{\text{baseline}}|$. Two perturbations are reported: a cross-spectrum permutation in which y_i is shuffled across the 64 spectra, and a per-spectrum mean replacement in which y_i is replaced by the spectrum-wise mean. Both perturbations leave the centre output unchanged at the level reported in Table 8, more than 400 times below the 10^{-4} audit threshold, so the held-out gain in Section 5.2 cannot be attributed to leakage of y_i into $\hat{x}_i$.

Table 8. Blindspot integrity audit. Each row reports the central-pixel deviation $|\hat{x}_i^{\text{perturbed}} - \hat{x}_i^{\text{baseline}}|$ aggregated over three test-pixel positions and 64 held-out spectra under a non-information-preserving perturbation of y_i . Both deviations are more than 400 times below the 10^{-4} audit threshold.

Perturbation of y_i	Pixels tested	Spectra	Max $ \Delta $
Cross-spectrum permutation	3	64	1.49×10^{-7}
Per-spectrum mean replacement	3	64	2.38×10^{-7}

C. VALIDATION-TUNED CONTROLS AND CONTEXT-VARIANCE ESTIMATORS

This appendix records the validation grids for the non-neural reference methods of Table 3, the per-estimator γ measurements summarized in Section 5.1, and the model-free predictor configurations used to obtain them.

Baseline tuning grids.—Table 9 lists the validation grid searched for each non-neural reference method. Within each family the configuration that maximizes the linear-form reconstruction S/N of Equation 11 on the validation split at a fixed validation noise seed is the configuration evaluated on the held-out test split in Table 3 and Table 10; the same selection rule is applied to every family so the two test-split tables are internally comparable.

Table 9. Validation grids searched for the non-neural reference methods. The selection rule for every family is the validation-split reconstruction S/N maximizer at a fixed validation noise seed, before any held-out evaluation. The realized winning rank for the PCA-family rows is listed; the realized winning configurations for the smoother families correspond to the rows reported in Table 3 (Wavelet BayesShrink) and Table 10 (Savitzky–Golay, Gaussian smoothing, median filter).

Family	Grid searched	Selection rule
Savitzky–Golay	window $\in \{3, 5, 7, 9, 11\}$, order $\in \{1, 2, 3\}$	validation S/N maximizer
Gaussian smoothing	$\sigma \in \{1, 2, 3, 5, 7\}$ pixels	validation S/N maximizer
Median filter	window $\in \{3, 5, 7\}$ pixels	validation S/N maximizer
Wavelet BayesShrink	db4, level $\in \{3, 4, 5\}$	validation S/N maximizer
Noisy-PCA	$k \in \{16, 32, 64, 128, 256, 512\}$, noisy-train fit	validation S/N maximizer ($k = 16$)
Clean-template PCA oracle	$k \in \{16, 32, 64, 128\}$, clean-template fit	validation S/N maximizer ($k = 16$)

Reference-baseline reconstruction quality.—Table 10 resolves the validation-selected baselines onto the same held-out test split, augmenting the four reference rows of Table 3 with median and bracket statistics and Ca II EW error, so that the line-preservation and reconstruction columns are measured against the same baseline rows.

Table 10. Validation-selected baseline denoisers on the held-out test split. Mean and median S/N columns report linear-form per-spectrum aggregations of Equation 11; the corresponding dB-form per-spectrum mean for the noisy input row is -4.26 dB. The Ca II column reports the across-spectra mean/median of the per-spectrum triplet equivalent-width relative error and is provided so that the line-preservation and reconstruction columns are scored against the same baseline rows. Table 5 reports per-line medians for the noisy and blindspot rows and uses a different aggregation, so the two presentations are not numerically interchangeable but report consistent orderings.

Reference	Mean S/N	Median S/N [5%,95%]	Ca II EW error	Polarity flip
Noisy input	7.8	7.8 [3.02,14.5]	2.50/0.817	0.277
Savitzky–Golay	21.3	19.8 [9.57,38.7]	1.58/0.578	0.118
Gaussian smoothing	26.4	23.4 [13.1,48.6]	1.41/0.653	0.0819
Median filter	16.1	15.2 [6.62,28.8]	1.83/0.633	0.164
Wavelet BayesShrink	27.2	24.9 [13.3,49.5]	1.31/0.686	0.0781
Noisy-PCA, $k = 16$	54.4	51.4 [26.6,91.1]	0.990/0.333	0.0182
Clean-template PCA oracle, $k = 16$	56.0	53.3 [27.3,93.3]	0.930/0.314	0.0171

Per-estimator γ measurements.—Table 11 reports the model-free measurement of the context-variance ratio $\gamma = \tau^2/\sigma^2$ summarized in Table 2. The three predictor families and their tuned configurations are: local cubic interpolation (window radius 3 pixels, selected over $\{3, 5, 8, 16, 32\}$); context-only PCA reconstruction with k leading components (selected $k = 256$ over $\{16, 32, 64, 128, 256\}$); and local ridge regression on a context Gaussian-process surrogate (window radius 32 pixels, ridge regularization $\alpha = 10^{-8}$, selected over a 5×5 grid of radius $\in \{3, 5, 8, 16, 32\}$ and $\alpha \in \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 1\}$). Each family is tuned on the validation split by minimum residual variance and evaluated on the held-out test split.

D. DISTRIBUTIONAL CHECKS OF RECONSTRUCTION AND LINE PRESERVATION

This appendix checks that the reconstruction-S/N gain of Section 5.2 and the line-preservation gain of Section 5.4 are distributed across the held-out input-S/N range and across diagnostic lines, rather than driven by a small subset of

Table 11. Model-free measurements of $\gamma = \tau^2/\sigma^2$ on held-out clean spectra. Ca II, O₂, and TiO entries summarize the corresponding diagnostic windows. The median-of-three and worst-of-three rows are the aggregations promoted to the main-text Table 2.

Predictor or aggregate	Median γ	95th pct. γ	Ca II 8542	O ₂ A-band	TiO 7650
Local cubic interpolation	6.37×10^{-5}	9.90×10^{-4}	6.70×10^{-3}	4.89×10^{-5}	9.01×10^{-5}
Context-only PCA	2.61×10^{-4}	5.75×10^{-4}	1.68×10^{-3}	1.84×10^{-4}	3.91×10^{-4}
Local ridge regression	9.32×10^{-7}	6.67×10^{-5}	3.35×10^{-4}	9.21×10^{-7}	1.22×10^{-6}
Median of three	2.50×10^{-5}	8.80×10^{-5}	6.94×10^{-4}	1.97×10^{-5}	2.60×10^{-5}
Worst of three families	2.81×10^{-4}	1.00×10^{-3}	6.70×10^{-3}	2.20×10^{-4}	3.91×10^{-4}

spectra. The polarity-flip threshold $0.02 m_s$ is fixed across all rows of Table 5 and Table 10; sensitivity to alternative thresholds is recorded as future work.

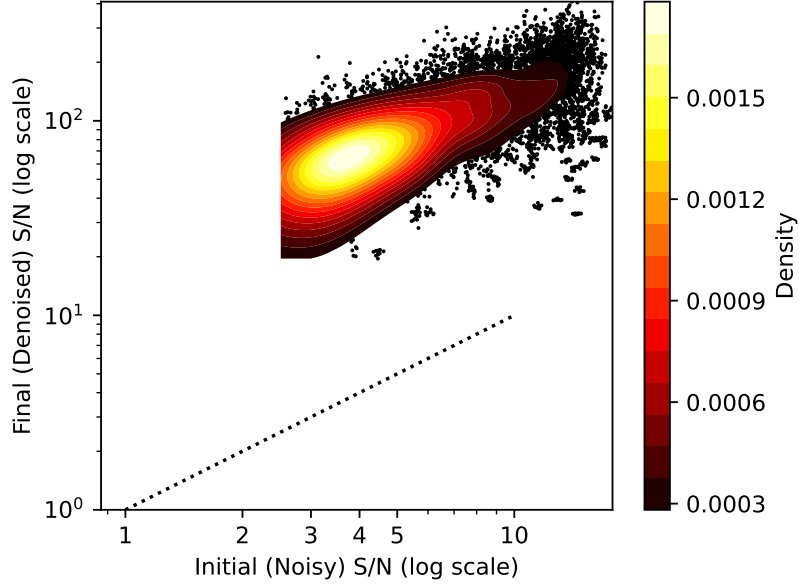


Figure 4. Joint density of per-spectrum input (noisy) and output (denoised) signal-to-noise ratio on the held-out test set, on log-log axes. The two-dimensional density (hot colormap; KDE) is overlaid with low-density outliers as black points; the dotted diagonal marks the $y=x$ identity. The bulk lies above the diagonal across the full input-S/N decade, so the reconstruction gain is approximately log-uniform in input S/N rather than driven by any single regime.

Figure 4 resolves the held-out aggregate reconstruction S/N of Section 5.2 into the joint distribution of per-spectrum input and output values, while Figure 5 resolves the line-preservation result of Section 5.4 into the dispersion of the EW relative error across diagnostic features as input S/N varies. Together they show that neither the S/N gain nor the EW-error reduction is dominated by a narrow input-S/N subset: the input-output S/N density lies above the identity across the full input-S/N decade, and the dispersion of the EW relative error after denoising is reduced over the entire diagnostic feature set rather than for a single Ca II core or a single S/N bin.

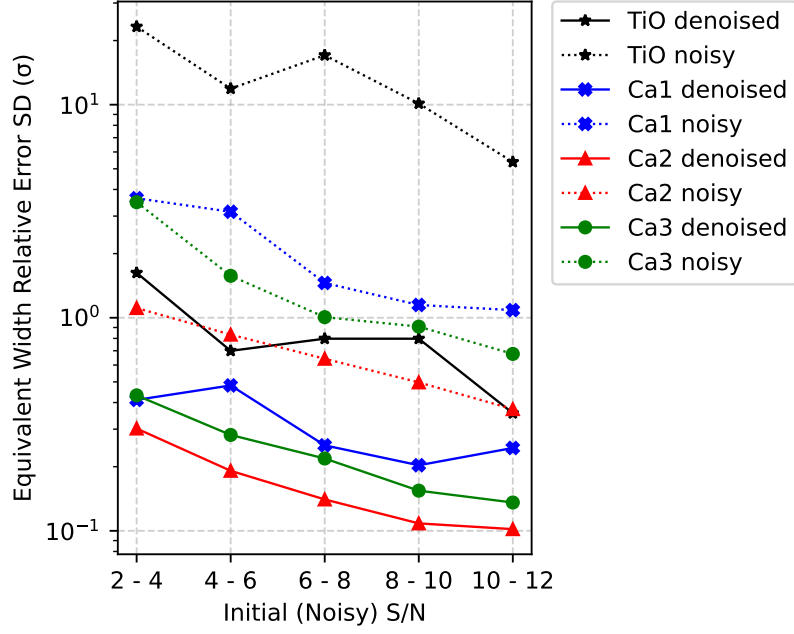


Figure 5. Equivalent-width preservation across diagnostic lines and input S/N. Standard deviation of the EW relative error as a function of input S/N for the TiO band near 7650 Å and the three Ca II triplet lines (solid: denoised; dotted: noisy). The denoised dispersion is reduced by an order of magnitude in the lowest-S/N bin and the gain narrows monotonically as input S/N rises, so the recovery is most useful where the noisy measurement is least informative; lines with intrinsically larger EW recover more cleanly than the weaker TiO band, consistent with line strength governing recoverability.